



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Tutorials on Java and Strings



Learning Outcome:

Talk about principles, concepts, and characteristics of general-purpose computer programming language in spoken and written situations.

Ask your partner:

- Do you consider Java is necessary when using your computer? Why?
- As a programmer, why is it important to learn and identify the syntax of certain programming tools?

Can Do Statements:

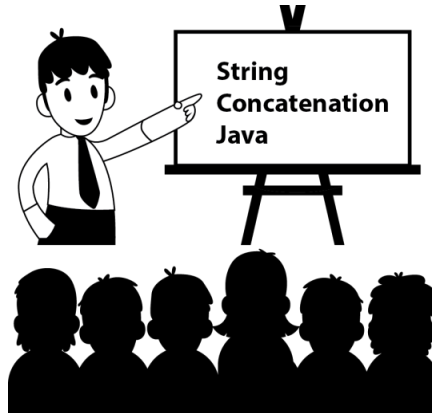
- I can say what a string in Java is.
- I can talk about the syntax to use in the string data type.
- I can say how concatenation works in the string data type.





Let's Explore!

Let's talk about Java.



Instructions: Read the text below.

Teacher: Let's talk about strings in java. Does anybody know what a string is?

Student: I read that a string is a sequence of characters considered as objects in Java.

Teacher: Exactly! Don't forget that strings in java are immutable. This means that once you've created a string you can never change it. Before we continue, do you remember what syntax is?

Student: When you declare a string you need to know about syntax. It is basically a set of rules -with variables and types of characters- that define and state how a Java program is written.

Teacher: Good! Let's continue. It's very common to create a new string with a string literal, which is a group of characters that are enclosed in double quotes (" "). Remember that = " " is equal to an empty space. Then, you reassign the keyword to the new string + = "My name is Bob." Remember that we can add strings together, so we use concatenation (+) at the end of each keyword to relate the sequences.

Student: How efficient is this method?

Teacher: I have to say that this method is quite inefficient because it starts with a single character and an empty space = " ". This will be an empty space forever. Every time we create a new string and reassign the keyword, we're creating a new string made of the previous one.

Student: How detrimental is this method?

Teacher: If you're writing a big program, this method could slow it down or use a lot of memory, so, to get around this, we should use another method. We should create a string with content. Let's say "new StringBuilder" and then we reassign a keyword. In this way we're not creating a new string but we're modifying the content of the existent one.

Student: What about lower or upper case in Java? How should I type?

Teacher: Well, to make lower or upper case in java just type in toUpperCase or toLowerCase.



Let's Practice!

Teacher describes the string concept in Java.

Instructions: Proofread the sentences. Correct the mistakes.

Student: I known a string be a sequence of characters or code.

Teacher: Exactly! Don't forget that strings in java is immutable. This mean that once you've created a string you can never change it. Before we continues, do you remember what syntax is?



Let's Practice!

Those are some of the questions teacher answered during the class.

Instructions: Use the words from the Word Bank to complete each sentence.

Word Bank

How efficient / easy / inefficient / How / be /
difficult is / syntax / is / simple is to

1. _____ this method?
2. How _____ can _____ to start with the single character and an empty space?
3. _____ easy is to create new strings based on a previous one?
4. How _____ to understand what _____ in Java is?
5. How _____ memorize special characters such as backlash or double quote?





Extension Activity

String relevance in Java programming language.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for extra information about string usages in Java programming language.
- Write notes and create a brochure to describe the string usages and main features.
- Share the brochure with your instructor and answer all his questions.

Self-Assessment

Yes

No

- I look for extra information about string usages in Java programming language.
- I create a brochure to describe the string usages and main features.
- I talk about string usages and main features in Java programming language.

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Target Words

- immutable** (adjective): something that doesn't change or vary
e.g. Don't forget that strings in java are immutable. This means that once you've created a string you can never change it.
- reassign** (verb): to put something again in a place
e.g. Then, you reassign the keyword to the new string.
- efficient** (adjective) : Someone or something competent that gets the intended results
e.g. How efficient is this method?
- Java** (noun): A computer programming language designed for Internet
e.g. I read that a string is a sequence of characters considered as an object in Java.
- concatenation** (noun): the act of linking together
e.g. We use concatenation (+) at the end of each keyword to relate the sequences.
- detrimental** (adjective): something that causes a damage in a process
e.g. How detrimental is this method?



Learning Points

Questions with How + Adjectives

We use questions with the **how + adjective** form to ask about the degree of strength of the adjective involved in the sentence.

To use questions with the **how+ adjective** form follow the next structure:

[How +adjective + verb+ subject]

How important is string in computer programming language?

Notice that in this type of questions we don't use an auxiliary verb such as Do/ Does. As this is a way of asking for the description of something or someone, we have to use **the verb to be**.

- **How efficient is** this method?
Well, it's not that efficient. We should try another one.
- **How important are** strings in Java?
They are quite important since Java uses them in its codes

Examples

- *How inefficient is this method?*
- *How detrimental is this method?*





Learning Points

String Methods

Strings, which are widely used in many programming languages, are a sequence of characters that Java uses as objects on its codes.

A string is a data type used in programming which is used to represent texts. It consists of sequences of characters or codes called escape codes. A string can be a word like "good" or a phrase "This is good". The data has to be enclosed in quotation marks to be recognized as a string and once it is created it can't be modified. Java string data is also considered as a class which is the reason why it has methods. Let's have a look at them.

String Length () Method

It is an accessor method used to get information of an object. Using the string length method you can get the number of characters, including spaces, the string object has.

Upper and Lower Case Methods

They are used to return the original string in upper case or lower case depending on the case.

Trim

This method is useful to get rid of the extra spaces of the string.

Replace ()

As its name implies, this method is helpful to replace an instance with another one

